

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10	U04						***** GATE VALVES *****			32	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED.
20											
20	U04		GATE	GA6B1	0.5	1.5	GATE 150# SCRD FNPT BS1400 LG2 BODY AND TRIM HW	0	0		
30	U04		GATE	G01AC41	2	8	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW	0	0	B1	ALL BOLTS ARE METRIC DIA.
40	U04	VA5	GATE	G01AC96	2	16	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW 2800MM EXT	0	0	FW3	VIKING JOHNSON COUPLINGS SHALL ONLY BE USED ON 'WF' FIREWATER SERVICE
50	U04	VA5	GATE	G01AC99	2	16	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW 3300MM EXT	0	0		
60	U04	VA5	GATE	G01AC95	2	16	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW 3700MM EXT	0	0	GA3	SPIRAL WOUND GASKETS SHALL ONLY BE USED BETWEEN CARBON STEEL FLANGES AND CARBON STEEL VALVES.
70											
70	U04						***** GLOBE VALVES *****				
80											
80	U04		GLOBE	L01AC41	2	6	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y HW	0	0	ND8	USE ONLY WHERE SPACE IS LIMITED AND WITH APPROVAL BY MATERIALS ENGINEER.
90	U04		GLOBE	L01AC42	8	8	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y GO	0	0	NX1	9" NOM SIZE (225MM O.D.) STUB FLANGES SHALL BE SUPPLIED C/W 8" NOM SIZE BACKING RINGS.
100											
100	U04						***** BUTTERFLY VALVES *****				
110											
110	U04		BUTTERFLY	T01AC61	10	24	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0	UDC	FOR USE WITH PRESSURE CONNECTIONS ONLY
120											
120	U04						***** PIPE *****			VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
130											
130	U04	PA5	PIPE	5713401	2	24	PIPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	0	0	PA5	EQUIVALENT HDPE PIPE TO NOMINAL PIPE SIZES SEE PIM-SU-5112
140	U04		PIPE	5713420	26	40	PIPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 (NON-STANDARD SIZE)	0	0		
150											
150	U04	PA5					***** FITTINGS *****				
160											
160	U04	ND8	ELBOW	5713801	2	12	ELBOW 90 DEG SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 EF	0	0		
170	U04		ELBOW	5713501	2	12	ELBOW 90 DEG ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	0	0		
180											
180	U04	ND8	ELBOW	5713811	14	24	BEND 90 DEG LONG RADIUS SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 EF	0	0		
190	U04		ELBOW	5713511	14	24	BEND 90 DEG LONG RADIUS SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	0	0		
200											
200	U04	ND8	ELBOW	5713802	2	12	ELBOW 45 DEG SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 EF	0	0		
210	U04		ELBOW	5713502	2	12	ELBOW 45 DEG ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	0	0		
220											
220	U04	ND8	ELBOW	5713810	14	24	BEND 45 DEG LONG RADIUS SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 EF	0	0		
230	U04		ELBOW	5713510	14	24	BEND 45 DEG LONG RADIUS SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	0	0		
240	U04		ELBOW	5713540	28	40	BEND 45 DEG LONG RADIUS SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 (NON-STANDARD SIZE)	0	0		
250											
250	U04	ND8	TEE	5713803	2	24	TEE EQUAL SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 EF	0	0		
260	U04		TEE	5713503	2	24	TEE EQUAL ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	0	0		
270	U04	ND8	TEE RED	5713804	2	18	TEE REDUCING SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 EF	4	24		
280	U04		TEE RED	5713504	2	18	TEE REDUCING ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	4	24		
290	U04		TEE RED	5713545	4	4	TEE REDUCING ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 (NON-STANDARD SIZE)	24	24		
300	U04		TEE RED	5713546	16	16	TEE REDUCING ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 (NON-STANDARD SIZE)	24	24		
310											
310	U04	ND8	CAP	5713805	2	24	CAP SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 EF	0	0		
320	U04		CAP	5713505	2	24	CAP ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	0	0		
330											
330	U04	ND8	RED CONC	5713806	2	14	REDUCER CONC SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 EF	3	16		
340	U04		RED CONC	5713506	2	14	REDUCER CONC ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	3	16		
350			RED CONC	5713535	6	8	REDUCER CONC ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 (NON-STANDARD SIZE)	12	16		
360			RED CONC	5713536	14	16	REDUCER CONC ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 (NON-STANDARD SIZE)	18	24		
370	U04	ND8	RED ECC	5713807	2	10	REDUCER ECC SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 EF	3	12		
380	U04		RED ECC	5713507	2	10	REDUCER ECC ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	3	12		
390											
390	U04		LATERAL	5713688	4	10	LATERAL 45 DEG SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	0	0		
400											
400	U04		Coupling	5713509	2	24	COUPLING ELECTRO FUSION SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	0	0		
410											
410	U04	UDC	SADDLE	5713515	2	2	PIPE SADDLE WITH GUNMETAL THREADED (NPT - FEMALE) OUTLET ADAPTOR	3	8		
420	U04	UDC	SADDLE	5713516	2	2	PIPE SADDLE WITH GUNMETAL THREADED (NPT - FEMALE) OUTLET ADAPTOR	10	24		
430											
430	U04	UDC	RED CONC	5544151	0.75	1.5	SWAGE CONC SCH XS/160 LTCS A420 WPL6 TBE (MNPT) GALV A153	2	2		
440											
440	U04	UDC	FITTING SCRD	5491325	0.5	1.5	BUSHING 3000# SCRD (MNPT/FNPT) HEX HEAD LTCS A350 LF2 CL.1 GALV A153	0.75	2		
450											
450	U04		Coupling	5713518	0.5	2	PIPE ADAPTOR WITH GUNMETAL THREADED (NPT-MALE) OUTLET	0	0		
460											

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT FUSION SEAL WELDING - NONE PIPE BENDS - NONE THREAD SEALANT - NONE ELECTRIC SOCKET WELDING OR BUTT WELDING AS APPLICABLE	CORR. ALLOW. 0 MM	DESIGN: DESIGN PRESSURE RANGE AS PER PIM-SU-5112-TCO CODE - SHALL BE ADVISED BY TCO SME FLUID CODES - PIM-SU-5112-TCO	MATERIAL: HDPE PE100 SDR 11 ID COLOR: BLUE FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 40 °C	12/16/2024	BY OZAC	U04	Page 1 of 2	150PE2

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460	U04	FW3	Coupling	5719243	3	24	MECHANICAL COUPLING VIKING JOHNSON DISMANTLING JOINT TYPE 150#FF	0	0	
470	U04						***** FLANGES *****			
480	U04	NX1	FLANGE	5713508	2	24	STUB FLG ELONGATED TYPE SDR 11 HD POLYETHYLENE C/W BACKING RING PE 100 WATER ASTM-D3350 / ISO 4427	0	0	
490	U04		BLIND	5621090	2	24	FLG BLIND FF 150# HDPE PE 100 WATER ASTM-D3350 / ISO 4427 WITH STEEL INSERT	0	0	
500	U04		FLANGE	5713541	28	40	STUB FLG ELONGATED TYPE SDR 11 HD POLYETHYLENE C/W BACKING RING PE 100 WATER ASTM-D3350 / ISO 4427 (NON-STANDARD SIZE)	0	0	
510	U04						**** PUDDLE FLANGE SPOOL ADAPTORS ****			
520		32	FLG ADAPTER	5597700	2	24	SPOOL - PUDDLE (ANCHOR) FLANGE WITH STUB FLANGE C/W BACKING RING AND WITH 2" PIPE SDR 11 HD PE 100 ASTM-D3350/ISO 4427	0	0	
530	U04						***** PUDDLE FLANGE SPOOLS *****			
540	U04		PUDDLE FLG	5597703	4	40	PUDDLE (ANCHOR) FLANGE WITH PIPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350/ ISO 4427 LENGTH 750MM	0	0	
550	U04						***** GASKETS *****			
560	U04		GASKET	5664201	2	24	GASKET 150# RF FLAT RING PTFE	0	0	
570	U04	GA3	GASKET	5673043	2	24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
580	U04	B1					***** BOLTING *****			
590	U04		BOLTING	5675768	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) W/ 2 NUTS + LTCS WASHERS	0	0	
600	U04		SPEC BLIND	5749582	2	14	SPECTACLE BLIND 150# RF LTCS A 516 GR.70 GALV A153	0	0	
610										

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		ALL SIZES - BUTT FUSION SEAL WELDING - NONE PIPE BENDS - NONE THREAD SEALANT - NONE ELECTRIC SOCKET WELDING OR BUTT WELDING AS APPLICABLE	CORR. ALLOW. 0 MM	DESIGN: DESIGN PRESSURE RANGE AS PER PIM-SU-5112-TCO CODE - SHALL BE ADVISED BY TCO SME FLUID CODES - PIM-SU-5112-TCO	MATERIAL: HDPE PE100 SDR 11 ID COLOR: BLUE FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 40 °C	12/16/2024	BY OZAC	U04	Page 2 of 2	150PE2